

ISO18000-3 mode 3 Air Interface Protocol Function Library

V1.3

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1. Instruction

The development kit supports the ANSI-C, ANSI-C++ and other languages to call the C function interface, such as delphi, VB, C#. The development kit cannot be used independently and must be combined with other RFID development kit. Before using, you must read “DLL Development Guide.doc” first, to make sure well understand the whole development process for SDK package.

2. Application Integration

This package consists of the following files:

Files	Description
rfidlib_aip_iso18000p3m3.dll	Dynamic link library, contains all API interface
rfidlib_aip_iso18000p3m3.lib	VC++ project, static link library
rfidlib_aip_iso18000p3m3.h	Header file of C/C++ project , the function definition
rfidlib_aip_iso18000p3m3.cs	C# class file
rfidlib_aip_iso18000p3m3.pas	Header file of Delphi function definition

Programming Language	Description
VC++	Copy rfidlib_aip_iso18000p3m3.lib and rfidlib_aip_iso18000p3m3.h to the project directory and add to the project. Copy rfidlib_aip_iso18000p3m3.dll to the exe file output directory.
C#	Copy rfidlib_aip_iso18000p3m3.cs to the project directory and add to the project. Copy rfidlib_aip_iso18000p3m3.dll to the exe file output directory.
Delphi	Copy rfidlib_aip_iso18000p3m3.pas to the project directory and add to the project. Copy rfidlib_aip_iso18000p3m3.dll to the exe file output directory.

3. Programming Interface

3.1. Overview

The programming interface is on the basis of `rfidlib_reader.dll`, define standard protocol command interface for belongs to ISO18000-3 mode 3, make it convenient for application developers use ISO18000-3 mode 3 protocol.

3.2. ISO18000p3m3_GetLibVersion

Definition	DWORD ISO18000p3m3_GetLibVersion(LPTSTR buf , DWORD nSize)		
Description	Get the version number of dynamic library		
Parameters	Name	Type	Remarks
	buf	LPTSTR	String buffer used for saving the version number
	nSize	DWORD	the max number of characters that can be saved in the buffer
Return (DWORD)	Return the number of characters that written in buffer. If return value equals “nSize”, indicate length of version more than the max number of characters that can be saved in buffer.		
Reference Code	<pre>char vbuf[128] ; DWORD written ; written = get_version(vbuf,128) ; if(written > 0) { /* show version string here */ printf(vbuf) ; } if(written == 128) { /* overflow,need to increase vbuf */ }</pre>		

3.3. ISO18000p3m3_CreateInvenParam

Definition	RFID_DN_HANDLE ISO18000p3m3_CreateInvenParam(RFID_DN_HANDLE hInvenParamSpecList, BYTE AntennaID ,BYTE Sel, BYTE Session,BYTE Q)			
Description	Create ISO18000-3 mode3 Inventory Parameters data nodes, used for RDR_TagInventory function			
Parameters	Name	Type	Direction	Remarks
	hInvenParamSpecList	RFID_DN_HANDLE	[in]	Handle of InventoryParameters list, created by RDR_CreateInvenParamSpecList
	AntennaID	BYTE	[in]	Antenna ID , 0 represents all antenna interfaces need to read tags at ISO18000-3 mode3 protocol , other Value means corresponding antenna ports
	Sel	BYTE	[in]	0x00=inventory object is all tags 0x01=inventory object is all tags 0x02=inventory object without SL identification 0x03=inventory object with SL identification
	Session	BYTE	[in]	Session type , 0-3 correspond to S0-S3
	Q	BYTE	[in]	Take the value range of 0-15, 0xFF is dynamic QValue
Return	If successful, returns the handle of data node.(RFID_DN_HANDLE)			
Reference Code				

3.4. ISO18000p3m3_SetInvenSelectParam

Definition	err_t ISO18000p3m3_SetInvenSelectParam(RFID_DN_HANDLE hIso18000p3m3InvenParam ,BYTE target ,BYTE action , BYTE memBank ,DWORD dwPointer,BYTE maskBits[], BYTE maskBitLen,BYTE truncate)			
Description	Set selection parameters of inventory			
Parameters	Name	Type	Direction	Remarks
	hIso18000p3m3 InvenParam t	RFID_DN_ HANDLE	[in]	The handle is created by ISO18000p3m3_CreateInvenParam
	target	BYTE	[in]	0x00: Modify target is Inventoried(S0) 0x02: Modify target is Inventoried(S2) 0x04: Modify target is SL
	action	BYTE	[in]	0x00: Match: Insert SL or change inventoried→A ; Not match: Remove SL or change inventoried→B ; 0x01: Match: Insert SL or change inventoried→A ; Not match: Don't do anything ; 0x02: Match: Don't do anything ; Not match: Remove SL or change inventoried→B 0x03: Match: Invert SL or (A→B, B→A) Not match: Don't do anything ; 0x04: Match: Remove SL or change inventoried→B Not match: Insert SL or change inventoried→A 0x05: Match: Remove SL or change inventoried→B Not match: Don't do anything ; 0x06: Match: Don't do anything ; Not match: Insert SL or change inventoried→A 0x07: Match: Don't do anything ; Not match: Invert SL or (A→B, B→A)

	memBank	BYTE	[in]	Storage region: 0x00: Reservation area ; 0x01: EPC arae ; 0x02: TID area ; 0x03: use area ;
	dwPointer	DWORD	[in]	Bit pointer
	maskBits	BYTE []	[in]	Data of mask bit, the buffer length is (maskBitLen + 7) / 8
	maskBitLen	BYTE	[in]	Mask bit length
	truncate	BYTE	[in]	Truncate or not
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.5. ISO18000p3m3_SetInvenReadParam

Definition	err_t ISO18000p3m3_SetInvenReadParam(RFID_DN_HANDLE hIso18000p3m3InvenParam ,BYTE MemBank ,DWORD WordPtr,BYTE WordCount) ;			
Description	Set read parameter of memory for inventory			
Parameters	Name	Type	Direction	Remarks
	hIso18000p3m3InvenParam	RFID_DN_HANDLE	[in]	This handle is created by ISO18000p3m3_CreateInvenParam
	MemBank	BYTE	[in]	Return air interface protocol type ID, refer to type ID list of air interface protocol of “DLL Development Guide.doc”
	WordPtr	DWORD	[in]	Return tag type ID, refer to Type ID list of tag of “DLL Development Guide.doc”
	WordCount	BYTE	[in]	Return ID of antenna interface
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.6. ISO18000p3m3_ParseTagDataReport

Definition	err_t ISO18000p3m3_ParseTagDataReport(RFID_DN_HANDLE hTagReport, DWORD *aip_id, DWORD *tag_id, DWORD *ant_id, DWORD *metaFlags, BYTE tagdata[], DWORD *tagDataLen) ;			
Description	Parsing inventory tag data information by ISO18000-3 mode 3			
Parameters	Name	Type	Direction	Remarks
	hTagReport	RFID_DN_HANDLE	[in]	Node handle of tag data information from RDR_GetTagDataReport
	aip_id	DWORD *	[out]	Return air interface protocol type ID, refer to type ID list of air interface protocol of “DLL Development Guide.doc”
	tag_id	DWORD *	[out]	Return tag type ID, refer to Type ID list of tag of “DLL Development Guide.doc”
	ant_id	DWORD *	[out]	Return antenna interface ID
	metaFlags	DWORD *	[out]	Data field identification, retention of 0 at present
	tagdata	BYTE []	[out]	When metaFlags=0 , the buffer consists of EPC only (Not include of PC and CRC)
	tagDataLen	DWORD *	[in,out]	In: the max length of “tagdata” buffer; Out: the bytes number written in “tagdata” buffer
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.7. ISO18000p3m3_Connect

Definition	err_t ISO18000p3m3_Connect(RFID_READER_HANDLE hr, DWORD tagType, BYTE epcBits[], DWORD epcBitsLen, BOOLEAN setPwd, DWORD accessPwd, RFID_TAG_HANDLE *ht) ;				
Description	Connect tag according to EPC code and create instance handle of corresponding tag				
Parameters	Name	Type	Direction	Remarks	
	hr	RFID_READER_HANDLE	[in]	Opened reader handle , created by RDR_Open	
	tagType	DWORD	[in]	Tag type ID, about the ID definition, please check the sixth part of “DLL Development Guide.doc”. If set 0, use reader default tag type	
	address_mode	BYTE	[in]	Addressing mode ;	
				Value	Description
				0	Non-address mode, uid can set empty.
				1	Address mode, need uid parameter value.
	2	Select mode , need uid parameter value.			
	uid	BYTE[]	[in]	Tag serial number, available only when address_mode is 1 or 2	
ht	RFID_TAG_HANDLE *	[out]	Connection is successful, return instance handle of tag		
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.				
Reference Code					

3.8. ISO18000p3m3_Read

Definition	err_t ISO18000p3m3_Read(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht ,BYTE memBank,DWORD startWord, BYTE wordCnt,BYTE wordBuffer[],DWORD *nSize)			
Description	Read data			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Driver Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO18000p3m3_connect
	memBank	BYTE	[in]	Storage area: 0x00: reservation area ; 0x01: EPC area ; 0x02: TID area ; 0x03: USER area ;
	startWord	DWORD	[in]	start word to read
	wordCnt	BYTE	[in]	Count of word, a word= two bytes
	wordBuffer	BYTE[]	[out]	Memory buffer, used to save data to read
	nSize	DWORD *	[in,out]	In: the max number of bytes that “wordBuffer” can write in Out: the number of bytes written
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.9. ISO18000p3m3_Write

Definition	err_t ISO18000p3m3_Write(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht , BYTE memBank, DWORD startWord, BYTE wordCnt, BYTE wordData[],DWORD nSize)			
Description	Write data			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Driver Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO18000p3m3_connect
	memBank	BYTE	[in]	Storage area: 0x00: reservation area ; 0x01: EPC area ; 0x02: TID area ; 0x03: USER area ;
	startWord	DWORD	[in]	Start word to write
	wordCnt	BYTE	[in]	Count of word, a word= two bytes
	wordData	BYTE[]	[in]	Memory buffer save data being write
	nSize	DWORD	[in]	The data length of buffer area
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.10. ISO18000p3m3_Lock

Definition	err_t ISO18000p3m3_Lock(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht ,WORD mask , WORD action) ;			
Description	Set access permissions of memory area			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Driver Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO18000p3m3_connect
	mask	WORD	[in]	mask of region set: bit0-1: USER area; bit2-3: TID area; bit4-5: EPC area; bit6-7: Access password area ; bit8-9: Kill password area; There are two mask of each area, set the two mask as b11 for the corresponding area to set permission.
	action	WORD	[in]	To do anction for memory region: bit0-1: Correspond with USER area; bit2-3: Correspond with TID area; bit4-5: Correspond with EPC area; bit6-7: Correspond with Access password area ; bit8-9: Correspond with Kill password area; There are two action types of each area, 1. USER 、TID、EPC three area types : b00: the area can be written in both open and safe state ; b01: the area can be written in both open and safe state ; b10: the area can be written in safe state ; b11: the area can not be written in any state; 2. Access and Kill password area: b00: the area can be read or written in both open and safe state ; b01: the area can be read or written in both open and safe state , also it can not be blocked ; b10: the area can be read or written in safe state ; b11: the area can not be read or written in any state;
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.11. ISO18000p3m3_Kill

Definition	err_t ISO18000p3m3_Kill(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht ,DWORD password , BYTE recommission) ;			
Description	Write data			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Driver Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO18000p3m3_connect
	password	DWORD	[in]	4 bytes of Kill password
	recommission	BYTE	[in]	Keeping unused, , must set 0
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.12. ISO18000p3m3_ReadPC

Definition	<code>err_t ISO18000p3m3_ReadPC(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht,WORD *pc);</code>			
Description	Read PC			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Driver Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO18000p3m3_connect
	pc	WORD *	[out]	Returns a 16-bit PC value
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				

3.13. ISO18000p3m3_WritePC

Definition	<code>err_t ISO18000p3m3_WritePC(RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht, WORD mask,WORD pc);</code>			
Description	Write PC			
Parameters	Name	Type	Direction	Remarks
	hr	RFID_READER_HANDLE	[in]	Reader Driver Handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO18000p3m3_connect
	mask	WORD	[in]	16 bit PC mask, if the corresponding mask bit is 1, modify the corresponding bit of the PC
	pc	WORD	[in]	16 bit PC value that needs to be modified
Return	0: Success Not 0: Failure, please check RFIDLIB API Error code table.			
Reference Code				